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Enhancing Needs-Based Segmentation: *Using Text Data Mining Framework for Attribute Discovery and Validation.* *An application in the field of Destination Branding.*

Thesis Defense for the Master's Degree in Data Science for Economics

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ALIBERTI

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MONTANELLI

December 2025



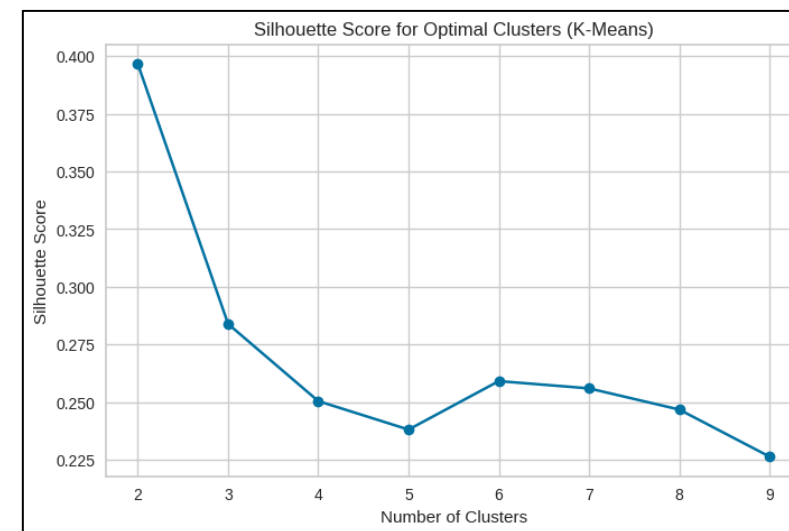
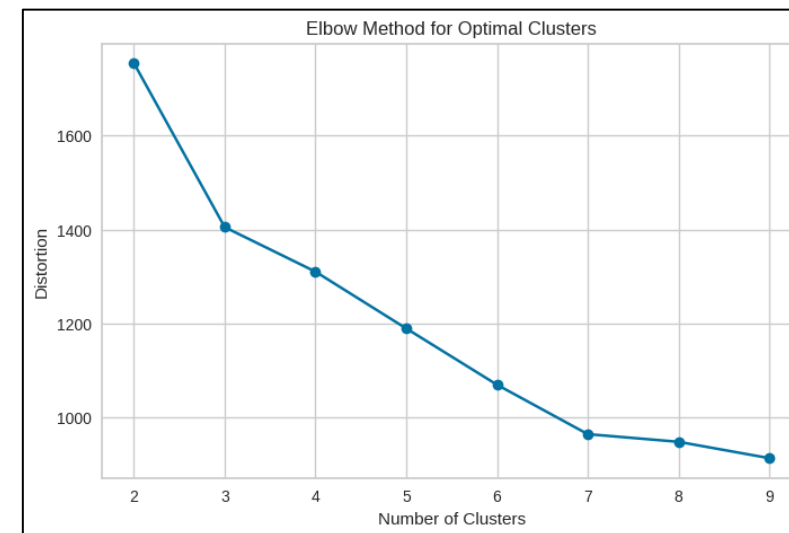


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MOTIVATION & PREVIOUS WORK



- ✓ As highlighted by Dibb & Simkin (2001), needs-based segmentation is highly dependent on the **quality and structure of the attributes** used.
- ✓ Cluster validation metrics showed **inconsistent indications**, suggesting that the underlying attributes were sub-optimal for producing stable and meaningful segments.
- ✓ The segmentation relied on K-means, a method not well suited for ordinal rankings, further contributing to instability.



Main Limitations

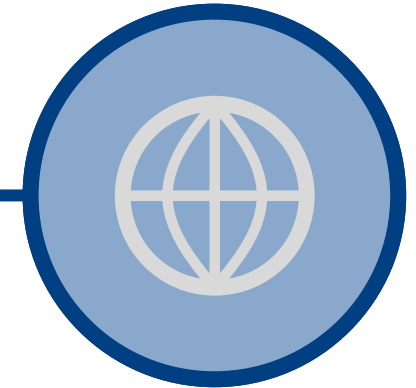


Main Limitations

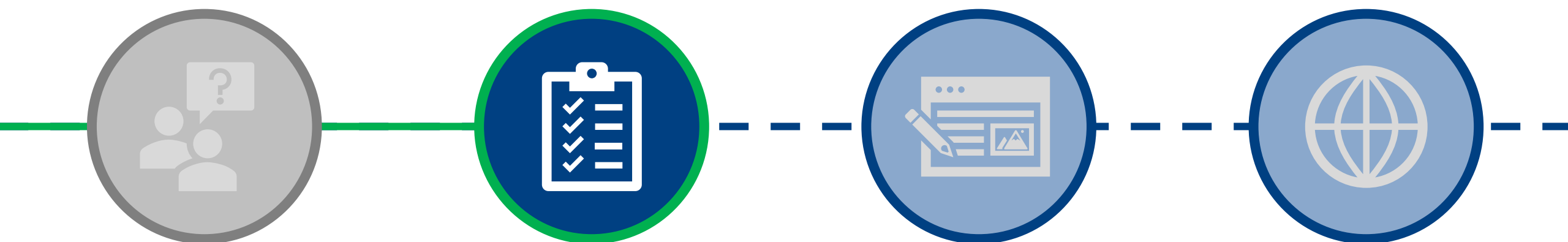


Qualitative interviews

- × Limited Questions.
- × Lack of genuine responses.
- × No recordings.



Main Limitations

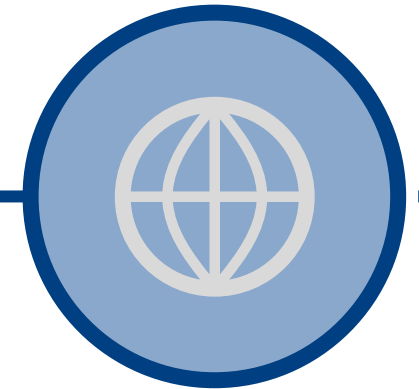


Set of Attributes

- × Too general attributes.
- × Ambiguity.

Previous Work Attributes
Accessibility/Public Transportation
Accessible Prices
Accommodation availability
Food
Historical background
Local events
Landscape
Availability of tourists' packs

Main Limitations



Quantitative survey

- × Small sample.
- × Avoidable questions.

Main Limitations



Language

- × Only English.
- × No native language.



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The background of the slide is a composite image. On the left, there is a tall, square stone tower with a triangular pediment and three arched windows on the upper level. A path leads through an arched opening at the base of the tower. To the right, a semi-transparent blue rectangle contains a faded image of a church with a prominent bell tower, set against a backdrop of green trees. The text 'RESEARCH QUESTION' is overlaid in white on this blue rectangle.

RESEARCH QUESTION

Question

*“How can **semi-automated text-analysis methods** be integrated into a **methodological framework** that identifies, evaluates, and selects **consumer-relevant product attributes** to improve needs-based market segmentation accuracy?”*

Objective

Design, implement, and test a **methodological framework** that combines **text mining with qualitative exploration and quantitative validation** into a single, coherent process.



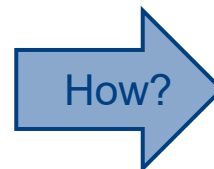
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METHODOLOGY & RESULTS

Main Idea



Main Idea

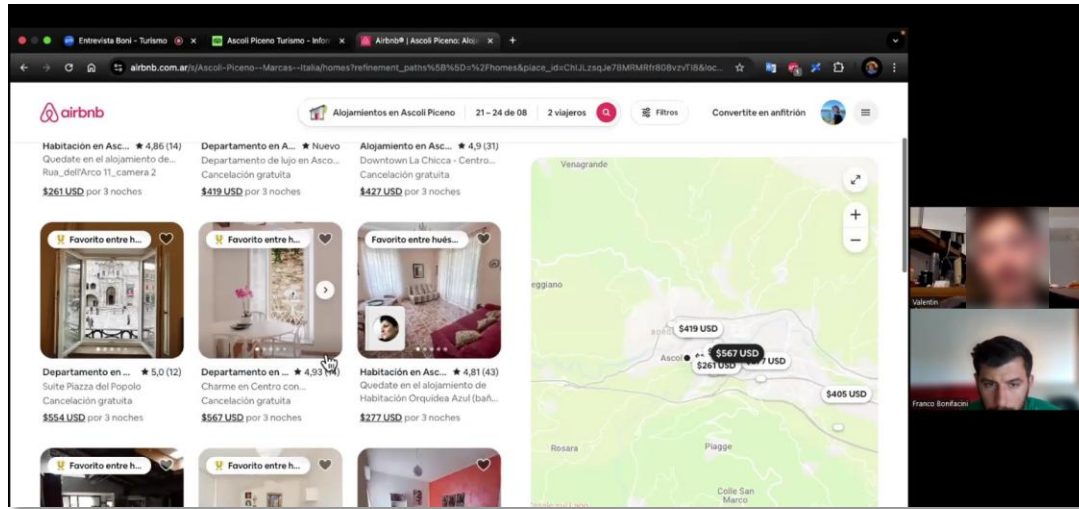


Methodological Steps





Methodology



- ✓ Online and recorded + Transcriptions.
- ✓ Flexible, open-ended questions + Free speaking
- ✓ Native language (*ideally*).
- ✓ Real-world experiences.
- ✓ Shared screen + Live planning.



10 interviewees

Manual identification of
attributes



TF-IDF &
BERTopic



LLM Prompt
Engineering



Results

Manual identification of attributes

Notes:

Interview 1 (Male, Italian, 24 years)

- Some important points, and insights:
 - "...look for ideas from other itineraries...". **ONLINE RECOMMENDATION**
 - "...it also depends on price, how to get there, if I can go by public transport or not, or if I can drive there...". **COST TO REACH, ACCESSIBILITY BY PUBLIC TRANSPORT**

Attributes:

#	Attribute	Frequency	#	Attribute	Frequency
1	accommodation type	9	23	overcrowded	3
2	online recommendations	9	24	restaurants availability	3
3	proximity to other destinations	9	25	accessible by public transport	2
4	local cuisine	8	26	accommodation quality	2
5	accessibility to touristic points	7	27	city's accessibility	2
6	cost to reach	7	28	genuine online recommendations	2
7	accommodation price	6	29	hiking trails	2
8	distance to reach	6	30	local friendliness	2
9	historical buildings	6	31	popular city	2
10	local events	6	32	variety of hours for transportation	2
11	accommodation services	5	33	access to mountains	1
12	availability of transport to destination	5	34	attraction availability	1
13	cultural immersion	5	35	availability of city tours	1
14	food's price	5	36	availability of travel agencies	1
15	proximity to popular cities	5	37	car friendly	1
16	accommodation location	4	38	destination size	1
17	historical places	4	39	events price	1
18	local prices	4	40	local experience	1
19	natural landscape	4	41	local restaurants	1
20	access to beaches	3	42	music events	1
21	information availability online	3	43	music events cost	1
22	museums	3	44	supermarket availability	1

Text Mining

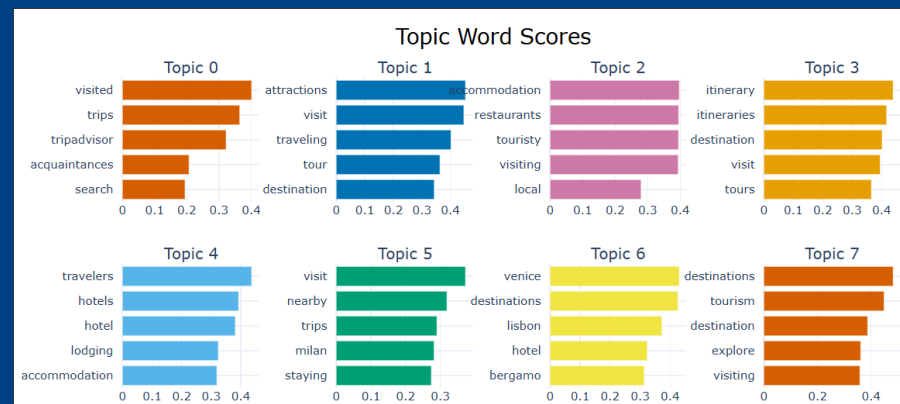
LLM Prompt Engineering

Text Mining

TF-IDF:



BERTopic:



LLM Prompt Engineering



Results

Manual identification of attributes

Text Mining

LLM Prompt Engineering

Original Output:

Attribute (English)	Frequency	Interviews where mentioned
Local food / gastronomy	10	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Architecture / historical buildings	9	1, 2, 3, 4, 5, 6, 7, 9, 10
Museums / art exhibitions	7	1, 2, 4, 6, 8, 9, 10
Cultural festivals / events	7	1, 4, 5, 6, 7, 8, 10
City atmosphere / vibe / livability	6	1, 2, 3, 6, 7, 8
Traditional restaurants / family-run eateries	6	3, 4, 6, 7, 9, 10
Historic centers / old towns	5	2, 4, 5, 9, 10
Local people friendliness / hospitality	5	3, 4, 6, 7, 10
Cleanliness of the city	4	3, 4, 6, 7
Artistic heritage / Renaissance art	4	2, 5, 8, 10
Squares / piazzas / public spaces	4	4, 7, 8, 10
Natural landscapes / vegetation / parks	3	4, 6, 7
Religious sites (churches, convents)	3	2, 6, 10
Guided tours / city tours	3	1, 5, 9
Nightlife / music festivals	3	1, 5, 8
Traditional markets / local markets	2	5, 8
Local crafts / authenticity / non-commercial atmosphere	2	4, 7
Street food / casual local food	2	9, 10
Festivals or parades (La Quintana)	2	8, 10
Scenic views / viewpoints / aesthetic beauty	2	6, 7

Refined Output:

Attribute
Quality of local cuisine
Presence of heritage architecture
Ambience of the historic center
Availability of museums and art venues
Frequency of cultural events
Quality of natural scenery
Friendliness of local residents
Cleanliness and orderliness of the city
Authenticity of local experience
Availability of guided tours and information on-site
Attractiveness of main squares and public spaces
Ease of access from other destinations
Cost and comfort of transport connections
Quality of accommodation options
Usefulness of online information
Scale and diversity of attractions
Affordability of the destination
Availability of visitor services



Results

Preliminary Set of Ten Attributes

Attribute
Distinctive local cuisine
Heritage ambience of the historic center
Availability of museums and art venues
Frequency of cultural events
Proximity to natural areas
Ease of reaching the city by public transport
Digital information availability
Accommodation type
Quality of accommodation options
Accessibility to touristic points



Methodology

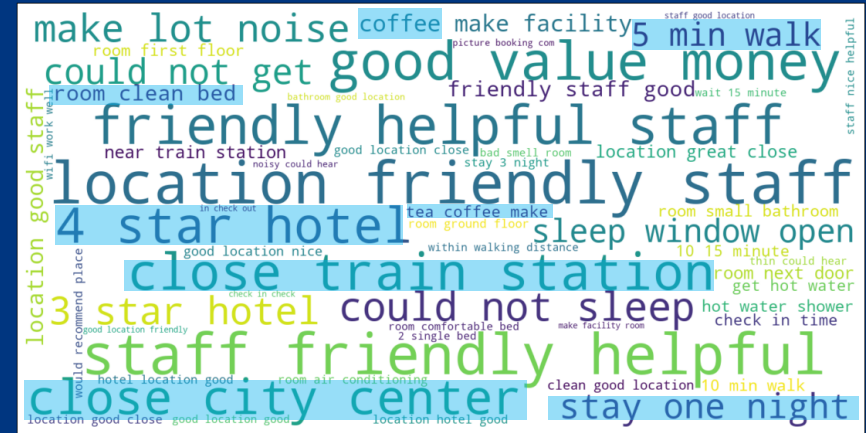
	Original Dataset	Optimization	Transformation	Frequency
	7,141,844	✓ Until upper quartile (75%) ✓ Cld2 detector → Only English (2,800,779) ✓ 15% sample	✓ Tokenization ✓ Lowercase ✓ Removal of stopwords and punctuation ✓ Lemmatization	✓ Unigrams ✓ Trigrams
	18,988	✓ Removal of "There are no comments available for this review"	✓ Tokenization ✓ Lowercase ✓ Removal of stopwords and punctuation ✓ Lemmatization	✓ Split into positive and negative reviews ✓ Unigrams ✓ Trigrams



Positive



Negative





Results

Preliminary Set of Attributes

Preliminary Set of 10 Attributes

Attributes from Reviews

Attributes	
	Distinctive local cuisine
	Heritage ambience of the historic center
	Availability of museums and art venues
	Frequency of cultural events
	Proximity to natural areas
	Ease of reaching the city by public transport
	Digital information availability
	Accommodation type
	Quality of accommodation options
	Accessibility to touristic points
NEW	Accommodation services
NEW	Availability of centrally located accommodation
NEW	Density of attractions within walking distance
NEW	Low tourist density

Methodology

Open-ended questions

PAST TRIPS

1. Tell us about a trip of yours that you were not satisfied with. What would you not do again, especially in the planning phase? (**OPEN-ENDED QUESTION**)

2. Which trip or vacation best represents your way of choosing how to travel? Tell us about the destination: how you chose it, why it reflects you, etc. (**OPEN-ENDED QUESTION**)

Decision-making steps

ORGANIZING A VACATION

3. Imagine organizing a vacation. Sketch a short story of your choice: what would be the 3–4 main scenes?

a. Scene 1 (**OPEN-ENDED QUESTION**)

b. Scene 2 (**OPEN-ENDED QUESTION**)

c. Scene 3 (**OPEN-ENDED QUESTION**)

d. Scene 4 (Optional) (**OPEN-ENDED QUESTION**)

MULTIPLE CHOICE

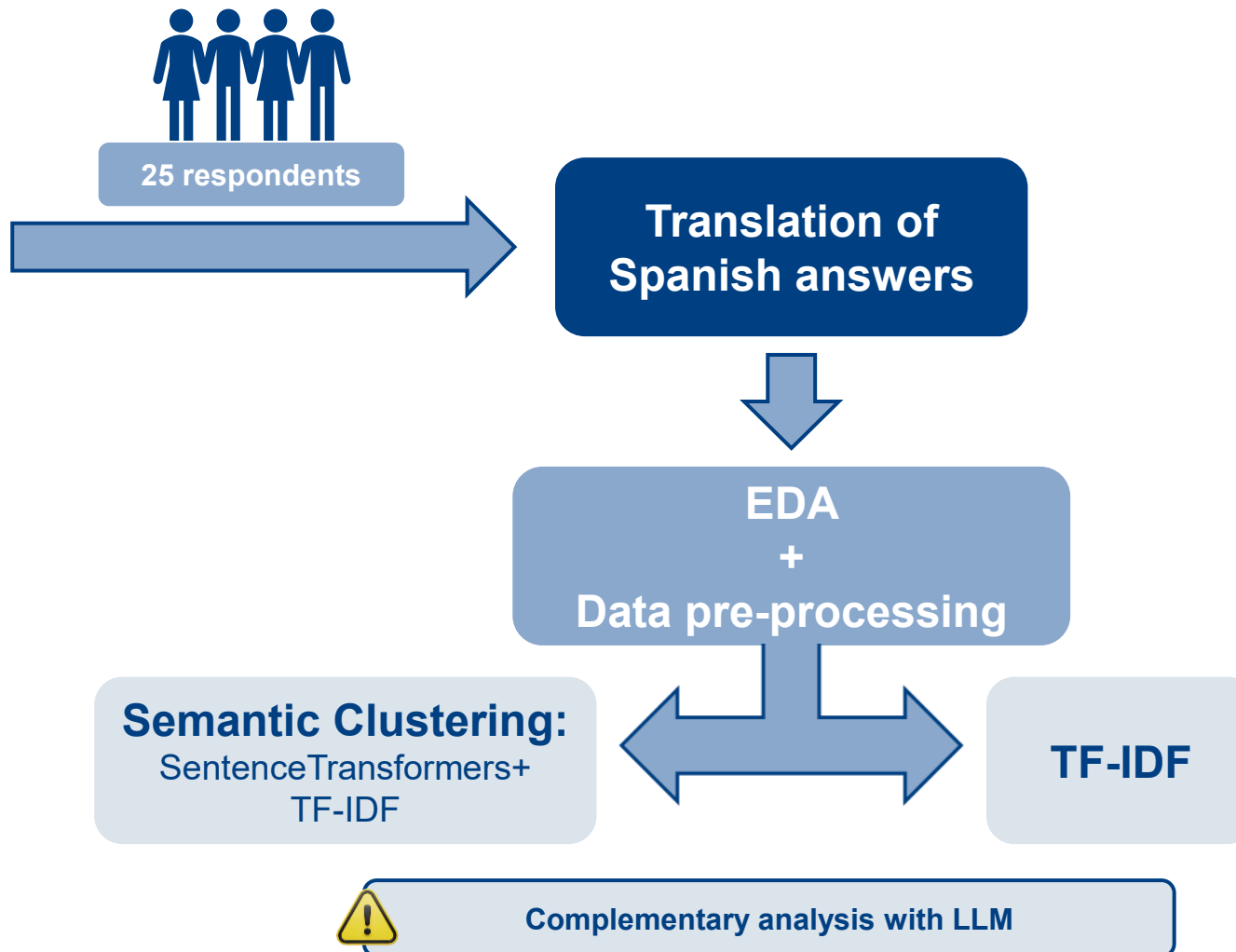
4. Which of these stories represents you the most?

a. I see an offer, check very little, and book the same day.

b. I compare 3–4 destinations, read reviews, and book after a few days.

c. I check flexibility/free cancellation and decide when I find the right one.

d. Others.



Questions were presented in English and Spanish



Complementary analysis with LLM



Results

New Takeaways

- ✓ Trips in which they visited **multiple destinations**, most often in Europe, a continent that enables visiting several cities within a single trip.
- ✓ Importance of **free-cancellation** for accommodations.
- ✓ Get informed from **reliable sources**, such as experienced travellers, review sites, and travel forums.
- ✓ Balance between **booking in advance** and leaving room for **spontaneous activities**.
- ✓ **Book tours** to get to know the city better.

Option A			Option B		
	bigram	tfidf		bigram	tfidf
0	price review	0.285714	0	free cancellation	0.473246
1	number review	0.142857	1	good rating	0.070357
2	base review	0.142857	2	number review	0.069786
3	short particular	0.045175	3	cancellation yes	0.051928
4	trip relatively	0.045175	4	cancellation good	0.040191
5	quantity review	0.045175	5	rating free	0.040191
6	relatively short	0.045175	6	well rating	0.032925
7	round trip	0.045175	7	cancellation help	0.031329
8	particular commitment	0.045175	8	sure free	0.031329
9	level quantity	0.045175	9	help decide	0.031329
10	consider level	0.045175	10	mind cancel	0.024494
11	commitment consider	0.045175	11	reassuring thing	0.024494
12	assume round	0.045175	12	thing change	0.024494

TF-IDF from reasons why respondents selected one option over the other, in the screen option question.



Preliminary Set of Attributes



Final Set of Attributes

Attributes
Distinctive local cuisine
Heritage ambience of the historic center
Availability of museums and art venues
Frequency of cultural events
Proximity to natural areas
Ease of reaching the city by public transport
Digital information availability
Accommodation type
Quality of accommodation options
Accessibility to touristic points
Accommodation services
Availability of centrally located accommodation
Density of attractions within walking distance
Low tourist density

Final Set of Attributes
Diversity of authentic local restaurants
Heritage ambience of the historic center
Availability of museums and art venues
Availability of centrally located accommodation
Proximity to natural areas or outdoor routes
City located outside major tourist circuits
Availability of tourist information (digital and on-site)
Accommodation services (flexible cancellation, digital check-in, amenities, etc)
Flexible scheduling of guided tours
Density of touristic attractions within walking distance



Removed Attributes

Preliminary Set of Attributes



Final Set of Attributes

Attributes
Distinctive local cuisine
Heritage ambience of the historic center
Availability of museums and art venues
Frequency of cultural events
Proximity to natural areas
Ease of reaching the city by public transport
Digital information availability
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Flexible scheduling of guided tours
Density of touristic attractions within walking distance



■ Removed Attributes

■ Refined Attributes

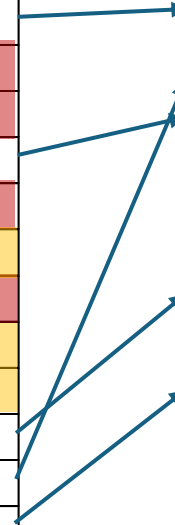
Preliminary Set of Attributes



Final Set of Attributes

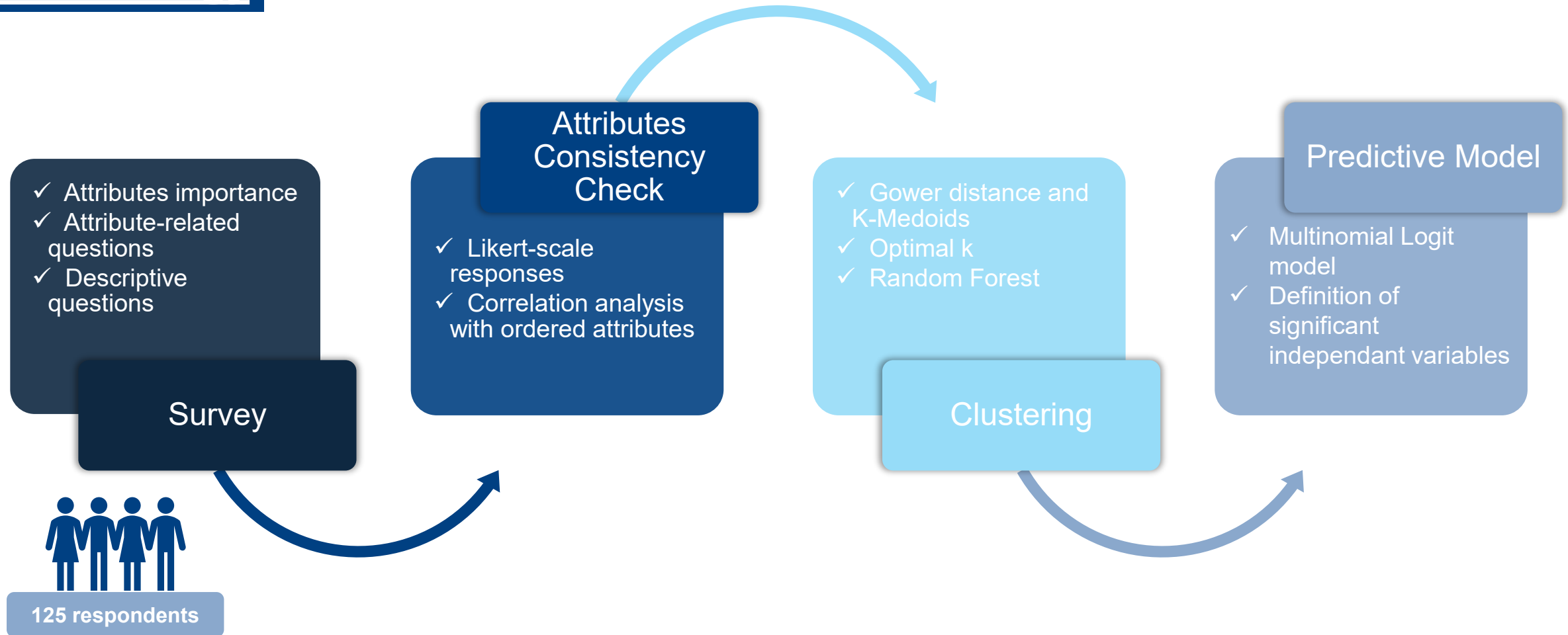
Attributes
Distinctive local cuisine
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Proximity to natural areas
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Final Set of Attributes
Diversity of authentic local restaurants
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Accommodation services (flexible cancellation, digital check-in, amenities, etc)
Flexible scheduling of guided tours
Density of touristic attractions within walking distance





Methodology





Results

Attributes Control

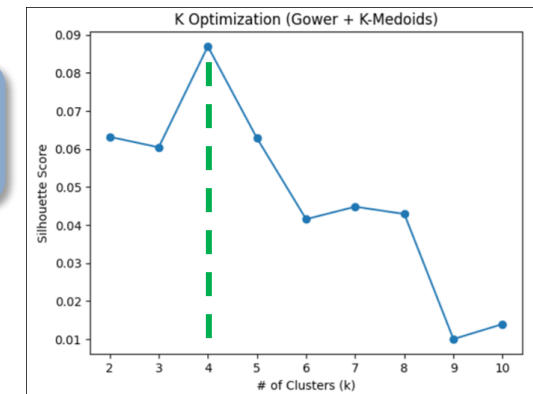
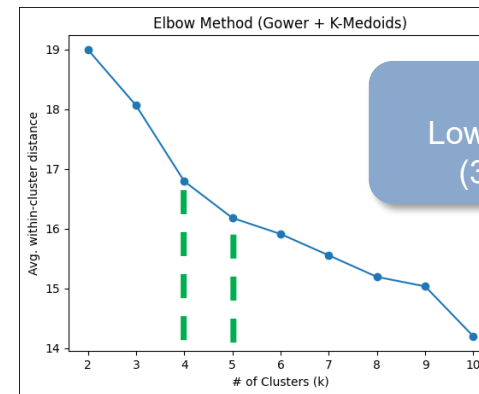
	Attribute	Spearman_rho	p_value	Significance
0	Diversity of authentic local restaurants	0.441	0.0000	****
1	Density of touristic attractions within walkin...	0.470	0.0000	****
2	City located outside major tourist circuits	0.208	0.0201	*
3	Availability of tourist information, both digi...	0.582	0.0000	****
4	Availability of centrally located accommodation	0.414	0.0000	****
5	Availability of museums and art venues	0.596	0.0000	****
6	Accommodation services, such as flexible cance...	0.451	0.0000	****
7	Heritage ambience of the historic center	0.526	0.0000	****
8	Proximity to natural areas or outdoor routes	0.529	0.0000	****
9	Flexible scheduling of guided tours	0.465	0.0000	****

Spearman
Correlation

Likert-scale
Answers

	1	2	3	4	5
att_local_restaurants_diversity	0	9	26	47	43
att_attractions_density	3	7	29	53	33
att_less_crowded_destinations	4	6	37	53	25
att_tourist_info_availability	3	11	29	53	29
att_central_accommodation	4	5	18	67	31
att_museums_art_availability	7	10	41	50	17
att_accommodation_services	4	5	27	58	31
att_heritage_ambience	4	6	20	55	40
att_natural_areas_proximity	1	6	40	55	23
att_guided_tours_flexibility	17	16	36	37	19

Clustering



	cluster	Population	1	2	3	4
Diversity of authentic local restaurants		3.92	3.29	4.17	5.61	3.10
Density of touristic attractions within walking distance		5.89	3.77	4.36	7.39	8.94
City located outside major tourist circuits		7.12	7.66	8.28	6.83	5.39
Availability of tourist information, both digital and on-site		6.41	5.71	7.42	5.39	6.77
Availability of centrally located accommodation		4.28	4.23	4.36	3.26	5.00
Availability of museums and art venues		5.04	6.66	4.14	5.65	3.81
Accommodation services, such as flexible cancellation, digital check-in, amenities, etc.		5.67	6.77	4.89	2.78	7.48
Heritage ambience of the historic center		3.23	3.77	2.33	5.26	2.16
Proximity to natural areas or outdoor routes		5.29	4.80	7.06	5.43	3.68
Flexible scheduling of guided tours		8.15	8.34	8.00	7.39	8.68

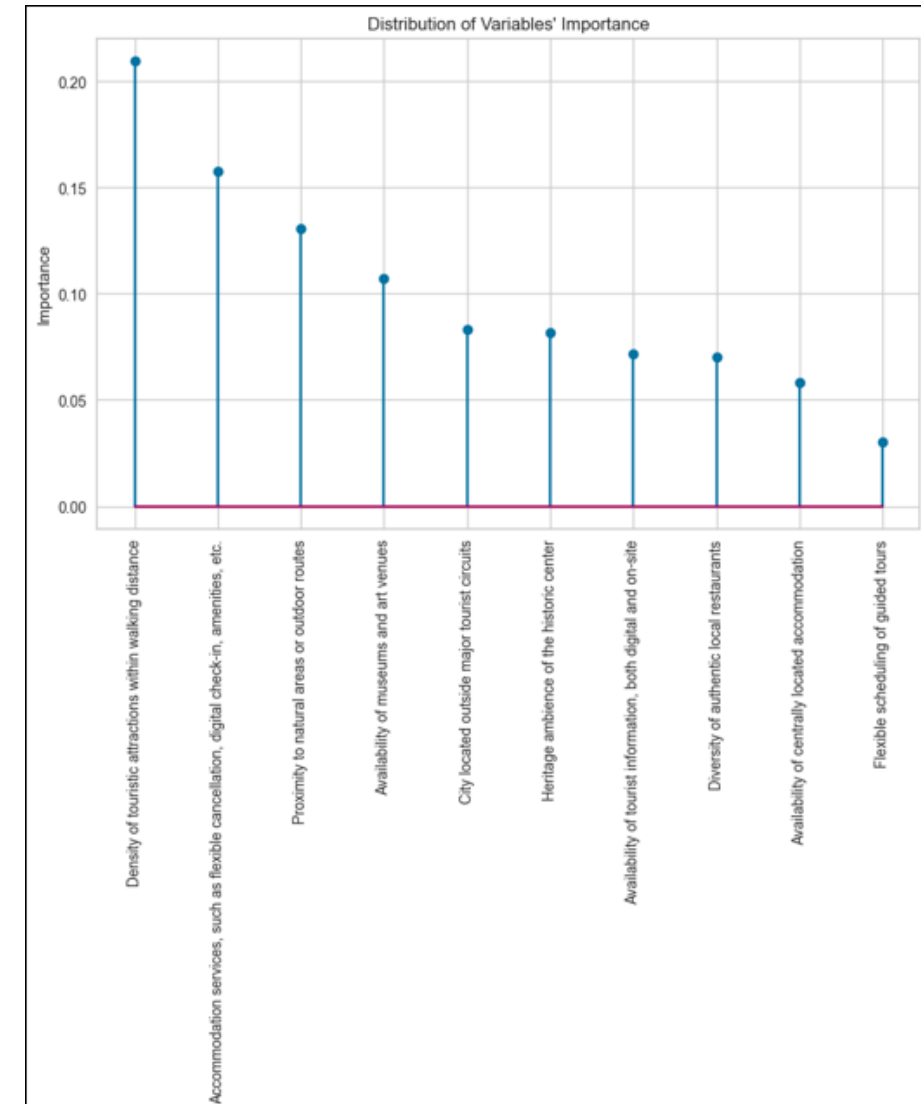


Results

Random Forest

	Importance
Density of touristic attractions within walking distance	0.21
Accommodation services, such as flexible cancellation, digital check-in, amenities, etc.	0.16
Proximity to natural areas or outdoor routes	0.13
Availability of museums and art venues	0.11
City located outside major tourist circuits	0.08
Heritage ambience of the historic center	0.08
Availability of tourist information, both digital and on-site	0.07
Diversity of authentic local restaurants	0.07
Availability of centrally located accommodation	0.06
Flexible scheduling of guided tours	0.03

Variable Importance in Random Forest Model





Results

Predictive Model

	Variable	VIF
19	const	96.218293
11	historical_architecture (architectural_interest)	3.385057
15	Beach (natural_environment_type)	2.622538
16	Natural parks (natural_environment_type)	2.590555
18	Lake/Rivers (natural_environment_type)	2.477505
17	Mountains (natural_environment_type)	2.428294
12	contemporary_cultural_interest (architectural_interest)	2.376164
7	city_tour_type	2.124081
2	residence	1.677343
13	personal_choice (restaurant_choice)	1.448941
0	age_group	1.434796
10	traditional_channels (accommodation_search)	1.321366
8	eating_preference	1.320544
5	less_crowded_willingness	1.296000
6	extra_travel_time	1.266397
14	social_influence (restaurant_choice)	1.255486
9	digital_platforms (accommodation_search)	1.231135
3	occupation	1.211654
4	define_accommodation	1.165311
1	gender	1.059941

Variance Inflation Factor (VIF) Scores

1st Model

MNLogit Regression Results			
Dep. Variable:	cluster	No. Observations:	125
Model:	MNLogit	Df Residuals:	65
Method:	MLE	Df Model:	57
Date:	Sat, 25 Oct 2025	Pseudo R-squ.:	0.2252
Time:	18:36:21	Log-Likelihood:	-132.89
converged:	False	LL-Null:	-171.53
Covariance Type:	nonrobust	LLR p-value:	0.03827

2nd Model

MNLogit Regression Results			
Dep. Variable:	cluster	No. Observations:	125
Model:	MNLogit	Df Residuals:	98
Method:	MLE	Df Model:	24
Date:	Sat, 25 Oct 2025	Pseudo R-squ.:	0.1104
Time:	18:36:21	Log-Likelihood:	-152.58
converged:	True	LL-Null:	-171.53
Covariance Type:	nonrobust	LLR p-value:	0.03563

cluster ~ age_group + gender + residence + occupation + city_tour_type
+ digital_platforms (accommodation_search)
+ traditional_channels (accommodation_search)
+ contemporary_cultural_interest (architectural_interest)



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LIMITATION & FUTURE WORK

Limitations:

- ✓ Final survey did not reach a **broader audience**.
- ✓ Missing some socio-demographic **variables**.

Limitations:

- ✓ Final survey did not reach a **broader audience**.
- ✓ Missing some socio-demographic **variables**.

Future work:

- ✓ Expanding **text sources** and collecting **multi-language responses**.
- ✓ Additional rounds of **qualitative** or **open-ended surveys**.
- ✓ Applying **latent-class** or **hierarchical models** → Deeper heterogeneity between tourist groups.
- ✓ Conducting a **conjoint** or **discrete-choice** experiment.
- ✓ Recommended to:
 - Adapting **behavioural questions** to destination-specific contexts.
 - **Predictive models** that focus directly on destination outcomes.



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CONCLUSION

- ✓ The proposed methodological framework appears promising in terms of **effectiveness and reliability**.
 - The interview process was redesigned to **reduce bias and improve reliability**.
 - The combination of qualitative interviews and automated text-based techniques **reduced subjectivity** and allowed the detection of **meaningful patterns**, previously not identified by a team of X analysts during the first attempted market research (PW).
 - The quantitative stage incorporated more **robust analytical tools**.
 - The final attributes showed **strong discriminant power** and **consistency**.
- ✓ **Contribution:**
 - **Strengthens** the early stages of attribute discovery.
 - Mixed methods can **improve** segmentation quality.
 - Offers **practical implications**.



THANK YOU!



ANNEX

Prompt used:

I'm going to send you several text files. I want you to read them carefully and then give me a detailed analysis that includes:

- *A list of specific tourism attributes (in English) found in each interview (not general ones like price or accommodation).*
- *The frequency of each attribute (across a total of 10 interviews).*
- *A table showing the attributes in English, their frequency, and which interviews they appear in.*

When I send you the files, confirm that you've read them, and then start the analysis. Don't write anything until I confirm that I've sent all the files.

Refinement prompt:

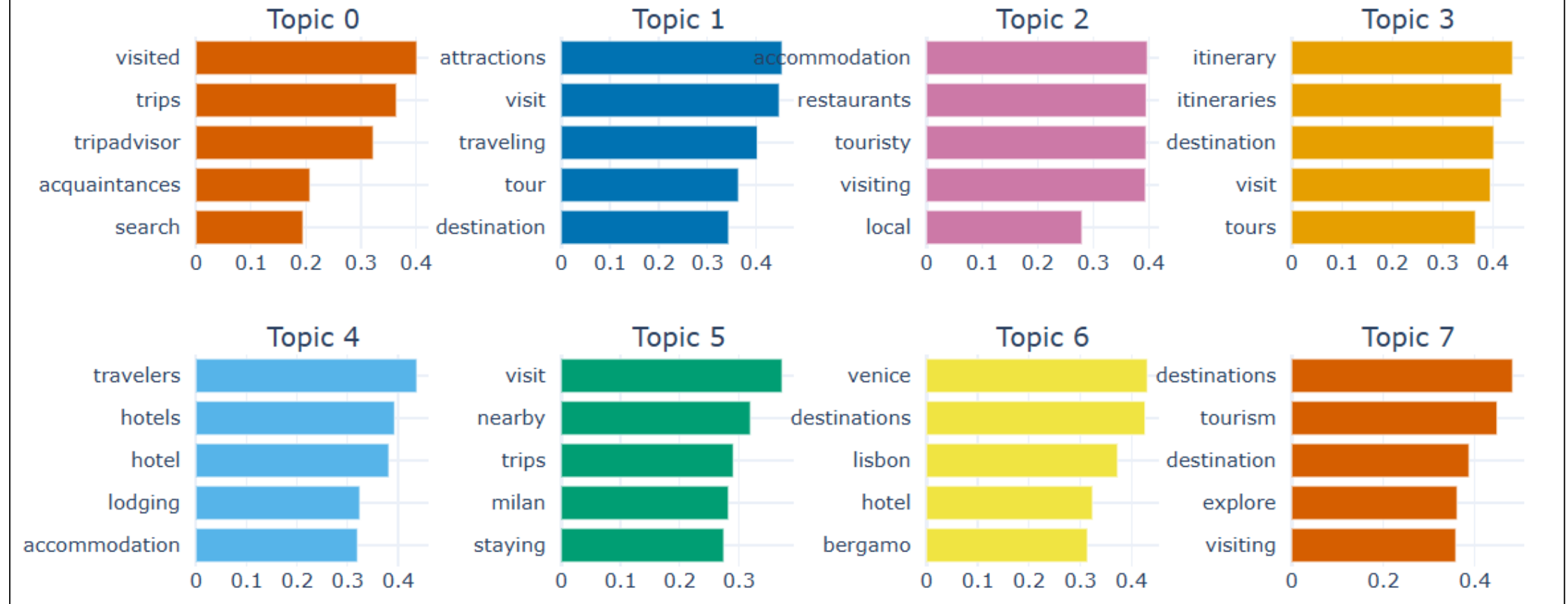
Continuing with this, it's important to understand that sometimes an attribute can be ambiguous for different respondents. For example, if I include "Historical architecture & monuments," it's not clear whether it refers to the availability of monuments, the presence of a historic center, etc.

That's why I need you to conclude with more detailed, clearly defined attributes that avoid ambiguity and don't overlap with other attributes.

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Topic Word Scores



Algorithm 1 Lemmatizing Reviews

Input: Data frame *airbnb_reviews_sample* with column *clean_tokens*

- 1: Initialize empty list *lemma_token_list* $\leftarrow []$
- 2: Create list *texts* $\leftarrow$ join tokens in *airbnb_reviews_sample[clean_tokens]* with spaces
- 3: **for** each *doc* in NLP pipeline applied to *texts* with batch size 2000 and *n_process* = 4 **do**
- 4: Extract *lemmas* $\leftarrow$ list of *token.lemma_* for each *token* in *doc*
- 5: Exclude tokens that are punctuation or spaces
- 6: Append *lemmas* to *lemma_token_list*
- 7: **end for**

Output: List *lemma_token_list* containing lemmatized tokens for each review

Algorithm 2 Computing TF-IDF Vectors

Input: List of documents *doc*

- 1: ▷ — Compute TF-IDF for unigrams —
- 2: Initialize *vectorizer_unigrams* $\leftarrow$ *TfidfVectorizer* with parameters:
 - *lowercase* = False
 - *token_pattern* = *r"(?u)\b\w+\b"*
 - *ngram_range* = (1,1)
 - *min_df* = 5, *max_df* = 0.8
 - *max_features* = 100,000
 - *dtype* = float32
- 3: *X_unigrams* $\leftarrow$ *vectorizer_unigrams* fit-transform on *doc*
- 4: *terms_unigrams* $\leftarrow$ *vectorizer_unigrams.get_feature_names_out()* ▷ —
Compute TF-IDF for trigrams —
- 5: Initialize *vectorizer_trigrams* $\leftarrow$ same parameters as above, but with *ngram_range* = (3,3)
- 6: *X_trigrams* $\leftarrow$ *vectorizer_trigrams.fit_transform(doc)*
- 7: *terms_trigrams* $\leftarrow$ *vectorizer_trigrams.get_feature_names_out()*

Output: TF-IDF matrices *X_unigrams* and *X_trigrams*, and their corresponding term lists

LLM PROMPT ENGINEERING ATTRIBUTES



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Results

Attribute	Observation
Centrally located accommodations.	Already in previous attributes
Accommodation within walking distance to key attractions.	Already in previous attributes
Accommodation's high quality (4/5 stars).	Already in previous attributes
Flexible booking options (free cancellation, easy date change).	Already in previous attributes
Accommodation near daily essentials (supermarket, pharmacy, ATM).	Already in previous attributes
Accommodation services (24h check-in or digital self check-in).	Already in previous attributes
Availability of authentic local restaurants with regional dishes.	Already in previous attributes
Restaurants with online or advance booking.	This is interesting, but should check if is relevant to include, taking into account that I already have something about restaurants.
Availability of local food markets/craft fairs.	Interesting, also saw this, but not mentioned by many tourists previously.
Availability of traditional, family-run restaurants.	This is also interesting, but again, related to local cuisine.
Possibility of cultural immersion (local rituals, traditions, or workshops).	Already in previous attributes
Availability of guided tours in different languages.	Clearly there is something about this matter, although was not frequent, with this approaches I reinforced the need of providing pre-fix activities so tourists avoid over-planning or FOMO (fearing not seeing something important)
Ready-made travel packages (include key attractions).	Can be combined with previous case
Local festivals.	Already in previous attributes and as mentioned before, this is stationary
Walkable historic districts.	Can be related to walking distance of accommodation and attractions.
Diversity of small local shops/artisans.	Related to other attributes already mentioned.
Availability of hiking/trekking routes.	Already in previous attributes
Proximity to other cities/villages.	Already in previous attributes
Proximity to natural areas.	Already in previous attributes
Ease of reaching the city by public transport.	Already in previous attributes
Availability of car rentals.	Already in previous attributes
Touristic sites reachable on foot.	Already in previous attributes
Availability of luggage storage near stations or airports.	This is really interesting attribute, yet it was not mentioned too much, and other attributes are more representative.

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